

Nifty 100 Financial Intelligence Platform

Analyst Guide

Screener Usage & Dashboard Navigation

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Covers: Stock Screener usage, Streamlit dashboard navigation, financial KPI engine, peer comparison, trend & sector analysis, capital allocation, valuation, cash flow intelligence, NLP insights, and automated PDF reports.

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1. Introduction & Platform Overview

The Nifty 100 Financial Intelligence Platform is a Python-based analytics suite for evaluating the constituents of the Nifty 100 index. It combines an ETL pipeline, a SQLite financial database, a financial KPI engine, an interactive Streamlit dashboard, and an automated ReportLab-based PDF reporting system into a single analyst workflow.

This guide is written for equity research analysts and platform operators who need to screen companies, compare peers, review trends, and generate shareable PDF reports without writing code. Command-line instructions are included for analysts who also want to run batch jobs directly.

1.1 What the platform does

- Loads and validates financial statement data for 92 Nifty 100 companies into a structured SQLite database.
- Computes 15 core financial KPIs (ROE, ROCE, margins, leverage, cash flow, CAGR metrics, and a composite quality score).
- Provides an interactive stock screener with six ready-made presets and CSV export.
- Supports peer group comparison with radar charts and median benchmarking across 11 peer groups.
- Analyzes 10-year trends, sector-level KPIs, capital allocation behaviour, valuation, and cash flow quality.
- Generates automated pros/cons narratives using a rule-based NLP parser with confidence scoring.
- Produces on-demand and batch PDF reports: company tearsheets, portfolio summaries, sector reports, screener exports, and peer group reports.

1.2 Who should use this guide

Audience	Primary use of the platform
Equity research analyst	Screen companies, review tearsheets, compare peers, export findings
Portfolio manager	Weekly portfolio summary PDF, sector rotation via Sector Analysis screen
Platform operator	Run batch report jobs, refresh the database, manage output archives
Reviewer / team lead	Validate report coverage and sign off on acceptance checklist deliverables

2. Getting Started (Installation & Setup)

The platform runs locally on Python 3 and requires no external services. Follow the steps below once to set up your environment.

Step 1 — Clone the repository

```
git clone https://github.com/Likithamurthy4/nifty100-financial-intelligence-platform.git
```

Step 2 — Move into the project directory

```
cd nifty100-financial-intelligence-platform
```

Step 3 — Create and activate a virtual environment

```
python -m venv venv
```

Windows: `venv\Scripts\activate` | macOS / Linux: `source venv/bin/activate`

Step 4 — Install dependencies

```
pip install -r requirements.txt
```

Core dependencies: Pandas, NumPy, Plotly, Streamlit, OpenPyXL, Matplotlib, ReportLab, Pytest, and Regex, running on a SQLite database at `db/nifty100.db`.

Step 5 — Launch the dashboard

```
streamlit run src/dashboard/app.py
```

This opens the Streamlit dashboard in your default browser, typically at `http://localhost:8501`. Keep the terminal window open while the dashboard is in use.

Step 6 — Verify the installation

```
pytest
```

The test suite covers financial ratio calculations, CAGR calculations, cash flow KPIs, ETL normalization, and data validation. All tests should pass before the platform is used for analysis.

3. Dashboard Navigation

The Streamlit dashboard is organized into eight screens, accessible from the left-hand sidebar. Each screen focuses on one stage of the analyst workflow, from a portfolio-wide overview down to company-level detail.

Screen	Purpose	Typical use
Home Dashboard	Portfolio-wide landing view	Quick health check across all 92 companies
Company Profile	Single-company deep dive	Review one company's KPIs and history before screening
Stock Screener	Filter companies by financial criteria	Build a shortlist using presets or custom filters
Peer Comparison	Compare a company against its peer group	Benchmark ROE, margins, and valuation vs peers
Trend Analysis	10-year historical trend charts	Assess revenue, profit, ROE/ROCE trajectory
Sector Analysis	Sector-level KPI aggregation	Compare sectors, spot sector rotation opportunities
Capital Allocation	Classify capital allocation behaviour	Identify reinvestors, distress signals, cash accumulators
Annual Reports	Access source annual report data	Cross-check screener output against filings

3.1 Home Dashboard

The landing screen after launch. It surfaces portfolio-level summary statistics so an analyst can spot outliers before drilling into individual companies or the screener.

3.2 Company Profile

Search or select a company to view its full KPI set, historical financials, and generated pros/cons narrative in one place. Use this screen to sanity-check a company before including it in a screener shortlist or requesting its tearsheet PDF.

3.3 Stock Screener

The primary filtering workspace, covered in full detail in Section 4. Apply preset or custom filters and export results to CSV.

3.4 Peer Comparison

Select a peer group (11 groups are configured) and compare member companies side by side using a comparison table and radar chart. See Section 6.

3.5 Trend Analysis

Plots 10-year revenue, net profit, ROE, and ROCE trends for one or more companies, with year-over-year growth annotations. See Section 7.

3.6 Sector Analysis

Aggregates KPIs by broad sector, shows sector median comparisons, and visualizes sector distribution across the Nifty 100 universe. The database currently spans 10 broad sectors (Communication Services, Consumer Discretionary, Consumer Staples, Energy, Financials, Healthcare, Industrials, Information Technology, Materials, and Real Estate).

3.7 Capital Allocation

Classifies each company's capital allocation pattern (Reinvestor, Deleverager, Distress Signal, Shareholder Returns, Cash Burner, Cash Accumulator, Balanced, or Insufficient Data) and tracks year-over-year pattern changes. See Section 8.

3.8 Annual Reports

Provides access to the underlying annual report source data used to populate the SQLite database, for analysts who need to trace a KPI back to its filing.

4. Using the Stock Screener

The Stock Screener is the fastest way to turn the 92-company universe into a shortlist. It supports both preset strategies and fully custom filter combinations, and every result set can be exported to CSV for offline work.

4.1 Filter reference

Filters are built from the platform's 15 core KPIs. Combine as many filters as needed; the screener applies them as an AND condition across companies.

KPI filter	What it measures
ROE	Return on Equity — profitability relative to shareholder equity
ROCE	Return on Capital Employed — efficiency of capital use
Net Profit Margin	Net profit as a percentage of revenue
Operating Profit Margin	Operating profit as a percentage of revenue
Debt-to-Equity	Leverage — total debt relative to shareholder equity
Interest Coverage	Ability to service interest expense from operating earnings
Asset Turnover	Revenue generated per unit of assets
Free Cash Flow	Operating cash flow after capital expenditure
Revenue CAGR	Compound annual growth rate of revenue
PAT CAGR	Compound annual growth rate of profit after tax
EPS CAGR	Compound annual growth rate of earnings per share
Composite Quality Score	Blended score combining profitability, leverage, and growth
Earnings Per Share (EPS)	Net profit attributable per share
Book Value Per Share	Net assets attributable per share
Dividend Payout Ratio	Proportion of profit distributed as dividends

4.2 Screener presets

Presets pre-load a sensible filter combination for a common analyst strategy. Select a preset from the dropdown, then optionally refine it with additional custom filters.

Preset	Typical filter emphasis
Quality	High ROE / ROCE, high Composite Quality Score, low leverage

Value	Attractive valuation multiples relative to sector peers
Growth	High Revenue, PAT, and EPS CAGR
Dividend	High and sustainable Dividend Payout Ratio
Debt-Free	Low or near-zero Debt-to-Equity
Turnaround	Improving margins or ROE off a depressed base

4.3 Reading and exporting results

- Results appear as a sortable table — click any column header to sort ascending or descending.
- Each row corresponds to one company; the composite quality score column is a useful default sort for a first pass.
- Use the Export to CSV control to save the current filtered result set for offline modelling or to attach to a note.
- To generate a polished, chart-based PDF of the same result set, use the Screener Output Report (Section 12) rather than the CSV export.

Tip: run a preset first, review the shortlist on the Company Profile screen, then layer on one or two custom filters (for example, a minimum Interest Coverage) to tighten the list before exporting.

5. Financial KPI Reference

All screens in the dashboard draw from the same KPI engine, so a metric means the same thing whether you see it on the Screener, a Company Profile, or a tearsheet PDF. The table below summarizes how each KPI should be read.

KPI	Higher is generally better	Watch-out
ROE	Yes	Can be inflated by high leverage
ROCE	Yes	More leverage-neutral than ROE
Net / Operating Margin	Yes	Compare within the same sector only
Debt-to-Equity	No	Zero is not always optimal for capital-intensive sectors
Interest Coverage	Yes	Below ~2x signals refinancing risk
Asset Turnover	Yes	Naturally low for asset-heavy sectors
Free Cash Flow	Yes	Check consistency across years, not one year alone
Revenue / PAT / EPS CAGR	Yes	Sensitive to the base year chosen
Composite Quality Score	Yes	A single blended score — always cross-check drivers
Dividend Payout Ratio	Context-dependent	Very high payout can signal limited reinvestment

Note: some companies have insufficient historical data for certain ratio or cash-flow calculations. The platform reports these as "Insufficient Data" or "N/A" rather than fabricating a value — treat these as data gaps, not zeros.

6. Peer Comparison Workflow

Use Peer Comparison when you need to judge a company relative to its closest competitors rather than the full 92-company universe.

- Step 1 — Open the Peer Comparison screen and choose one of the 11 configured peer groups.
- Step 2 — Select the companies within that group you want to compare (or leave all selected).
- Step 3 — Review the peer comparison table: KPIs are shown side by side with the peer median highlighted.
- Step 4 — Inspect the radar chart to see each company's shape across ROE, margins, leverage, and growth in one view.
- Step 5 — For a shareable version, request the Peer Group Report PDF (Section 12), which includes the same radar charts plus a side-by-side table for every peer group.

Reading the radar chart: a company whose shape sits outside the peer median line on profitability and inside it on leverage is showing peer-relative strength; the reverse pattern is a flag worth investigating on the Company Profile screen.

7. Trend & Sector Analysis

The Trend Analysis screen plots up to 10 years of history for revenue, net profit, ROE, and ROCE, with year-over-year growth call-outs and support for comparing multiple companies on the same chart. Use it to distinguish a genuine multi-year trend from a single good or bad year that a screener snapshot cannot show.

The Sector Analysis screen rolls the same KPIs up to the sector level, so you can compare, for example, Financials against Information Technology on median ROE, or see how a company's growth compares to its own sector rather than the whole index. Sector visualizations also show how companies are distributed within a sector by size and quality score.

8. Capital Allocation Classification

The Capital Allocation screen classifies how each company has deployed capital over time into one of eight patterns, and tracks whether that pattern is changing year over year.

Classification	What it indicates
Reinvestor	Capital is being redeployed into the business (capex, working capital)
Deleverager	Cash flow is being prioritized toward debt reduction
Distress Signal	Cash flow and leverage patterns consistent with financial stress
Shareholder Returns	Capital is being returned via dividends or buybacks
Cash Burner	Cash reserves are declining without a clear reinvestment pattern
Cash Accumulator	Cash is building on the balance sheet
Balanced	No single pattern dominates capital allocation
Insufficient Data	Historical data is too limited to classify confidently

Use the year-over-year pattern view to catch companies moving from Reinvestor toward Distress Signal — this transition is often visible before it shows up clearly in a single-period ROE or margin number.

9. Valuation Module

Valuation analysis benchmarks each company against sector-median multiples rather than absolute thresholds, since fair multiples vary widely by sector.

- FCF Yield — free cash flow relative to market value; higher generally signals better value.
- Sector Median P/E — the company's P/E compared against its own sector's median, not the whole index.
- EV/EBITDA — enterprise value relative to operating earnings, useful for comparing companies with different capital structures.
- P/B Ratio — price relative to book value, most relevant for asset-heavy or financial-sector companies.
- Overvaluation / Discount Detection — automatic flags where a company's multiple deviates materially from its sector median.
- Fair Valuation Classification — a summary label (overvalued, fairly valued, or discounted) built from the above signals.

10. Cash Flow Intelligence

Cash Flow Intelligence goes beyond the income statement to judge the quality and sustainability of a company's operating cash flow.

- CFO Quality Score and Classification — how closely operating cash flow tracks reported profit.
- CapEx Intensity and Classification — how much of operating cash flow is consumed by capital expenditure.
- Free Cash Flow CAGR and Conversion — growth and consistency of free cash flow generation.
- Distress Signal Detection — early-warning flags based on cash flow deterioration.
- Deleveraging Detection — identifies companies actively paying down debt from operating cash flow.
- Capital Allocation Integration — cash flow findings feed directly into the Section 8 classification.

In the current dataset, 13 companies are flagged with a distress alert; each alert includes the operating cash flow, financing cash flow, and latest net profit figures used to generate it, so the flag can be traced back to source numbers rather than treated as a black box.

Run `python src/analytics/cashflow_kpis.py` to refresh `output/cashflow_intelligence.xlsx` and `output/distress_alerts.csv`.

11. NLP-Based Financial Insights

The platform includes a rule-based NLP layer that reads free-text financial analysis (for example, management commentary or analyst notes), extracts structured figures such as CAGR and ROE using regex patterns, and generates automated pros and cons for each company.

- Each generated insight is tagged with a rule ID so an analyst can see which rule produced it.
- Every insight carries a confidence percentage — treat low-confidence signals as a starting point for manual review, not a final verdict.
- The current dataset produces 455 pros/cons records across 92 companies, with every company receiving at least one pro and one con.
- Structured output is written to `output/analysis_parsed.csv` and `output/pros_cons_generated.csv`, and any text that fails to parse is logged separately to `output/parse_failures.csv` for review.

Run `python src/demo_day29.py` for the parser and `python src/nlp/pros_cons_generator.py` for the pros/cons generator.

12. Automated PDF Reports

All PDF reports are generated with ReportLab and can be produced on-demand from the dashboard or in batch from the command line. This section summarizes each report type; Section 13 covers the exact commands.

#	Report	Contents	Frequency
8.1	Company Tearsheet (2-page)	P1: KPI tiles, 10yr revenue & profit bar, ROE/ROCE trend line; P2: Balance sheet composition stacked bar, cash flow detail, and the generated pros/cons narrative.	On-demand or batch
8.2	Portfolio Summary (1-page/co)	All 92 companies: name, sector, top 6 KPIs, 3yr trend arrow (up/down/right)	Weekly batch
8.3	Sector Report	Sector header page + all companies in sector; sector median KPI table; best/worst highlighted.	On-demand
8.4	Screener Output Report	Formatted PDF of screener results with charts, sorted by company score.	On-demand
8.5	Peer Group Report	Radar charts + side-by-side table for each of 11 peer groups	Weekly batch

12.1 Company Tearsheet (2-page)

Page 1 opens with KPI tiles for a fast read, followed by a 10-year revenue and net profit bar chart and an ROE/ROCE trend line. Page 2 shows balance sheet composition as a stacked bar, cash flow detail, and the generated pros/cons narrative. Output: reports/tearsheets/, one PDF per company, filename tearsheet.pdf within each company's folder.

Validated output: 92 companies, 92 PDFs, 0 skipped, 0 failed, 0 PDFs below 30 KB.

12.2 Portfolio Summary (1-page/co)

One page per company across all 92 companies in a single document, ordered alphabetically by ticker. Each page shows company name, sector, the six key financial KPIs, and a three-year trend indicator: an up arrow for improvement, a down arrow for decline, and a right arrow for changes within plus-or-minus 2%. Where historical comparison data is unavailable, the page shows N/A rather than a fabricated trend. Output: reports/portfolio/portfolio_summary_YYYYMMDD.pdf.

12.3 Sector Report

Opens with a sector header page, followed by every company in that sector, a sector median KPI table, and the best- and worst-performing companies in the sector highlighted. The Sprint 5 specification anticipated 11 sectors; the supplied database contains 10 distinct broad sectors, and a report is generated for each of the 10 sectors present in the source data. Output: reports/sector/.

13. Running Reports from the Command Line

Batch report generation is available for operators who need to refresh reports outside the dashboard, for example as a scheduled weekly or monthly job.

Company tearsheets

```
python src/reports/batch_tearsheets.py
```

Output: reports/tearsheets/

Sector reports

```
python src/reports/sector_report.py
```

Output: reports/sector/

Portfolio summary

```
python src/reports/portfolio_summary.py
```

Output: reports/portfolio/portfolio_summary.pdf

Valuation module

```
python src/demo_day26.py
```

Capital allocation

```
python src/demo_day32_part2.py
```

Output: output/capital_allocation.csv, output/pattern_changes.csv

Run the full test suite

```
pytest
```

Run the test suite after any data refresh or batch report run, before distributing PDFs to stakeholders.

14. Troubleshooting & FAQ

The dashboard won't start.

Confirm the virtual environment is activated and dependencies are installed (`pip install -r requirements.txt`), then re-run `streamlit run src/dashboard/app.py` from the project root.

A company's KPI shows N/A or Insufficient Data.

This is expected behaviour, not a bug — the platform will not fabricate a value when historical data is missing. For example, ATGL has insufficient historical financial data for some cash-flow and ratio calculations.

A sector report is missing for one of the 11 expected sectors.

The Sprint 5 specification anticipated 11 broad sectors; the supplied database contains 10. Reports are generated for all sectors that exist in the source data — this is a data-coverage note, not a generation failure.

The screener returns no results.

Check that filter thresholds are not mutually exclusive (for example, a very high ROE combined with a very low Debt-to-Equity may exclude every company in a capital-intensive sector). Loosen one filter at a time.

A generated PDF looks incomplete or is unusually small.

Re-run the relevant batch script; tearsheet generation validates that no PDF is produced below 30 KB, so a report failing that check should be regenerated rather than distributed.

Where do I find the raw source data behind a KPI?

Use the Annual Reports screen in the dashboard, or inspect `db/nifty100.db` directly with any SQLite client.

15. Appendix: Project Structure

Reference layout of the repository for analysts who need to locate a specific script or output file directly.

```
config/
data/
  raw/
  processed/
db/
  nifty100.db
docs/
  analyst_guide.pdf
  acceptance_checklist.pdf
  sprint5_retrospective.md
notebooks/
output/
  screener_output.xlsx
  peer_comparison.xlsx
  valuation_summary.xlsx
  valuation_flags.csv
  analysis_parsed.csv
  pros_cons_generated.csv
  cashflow_intelligence.xlsx
  distress_alerts.csv
  pattern_changes.csv
reports/
  radar_charts/
  tearsheets/           (92 PDFs, one per company)
  sector/               (one PDF per broad sector)
  portfolio/
    portfolio_summary.pdf
src/
  analytics/
    cashflow_kpis.py
  dashboard/
    app.py
  etl/
  nlp/
    parser.py
    pros_cons_generator.py
  reports/
    batch_tearsheets.py
    portfolio_summary.py
    sector_report.py
    tearsheet.py
  screener/
  utils/
tests/
```

requirements.txt
README.md

End of Analyst Guide. For deliverable sign-off status, see [docs/acceptance_checklist.pdf](#).